

PROJECT ASSIGNMENT 2

# Group10-PA2 Use Case Document

*Twelve user-goal use cases, UML system diagrams, and six alternative flow models*

FIFA.com and Chess.com

|               |                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|
| Course        | CSC13112 - UI/UX Design, FIT-HCMUS                                                                           |
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## 1. Scope and notation

This document specifies two proposed responsive web modules. UML use-case diagrams show actors outside named system boundaries and user-goal ellipses inside. The six concept diagrams in Section 8 are flow models, not use-case diagrams. Task specifications follow goal, precondition, subtasks, exceptions, context, and involved-party guidance (LN04 - Task Analysis, pp. 6, 39-40, 48-49).

## 2. Actors and external systems

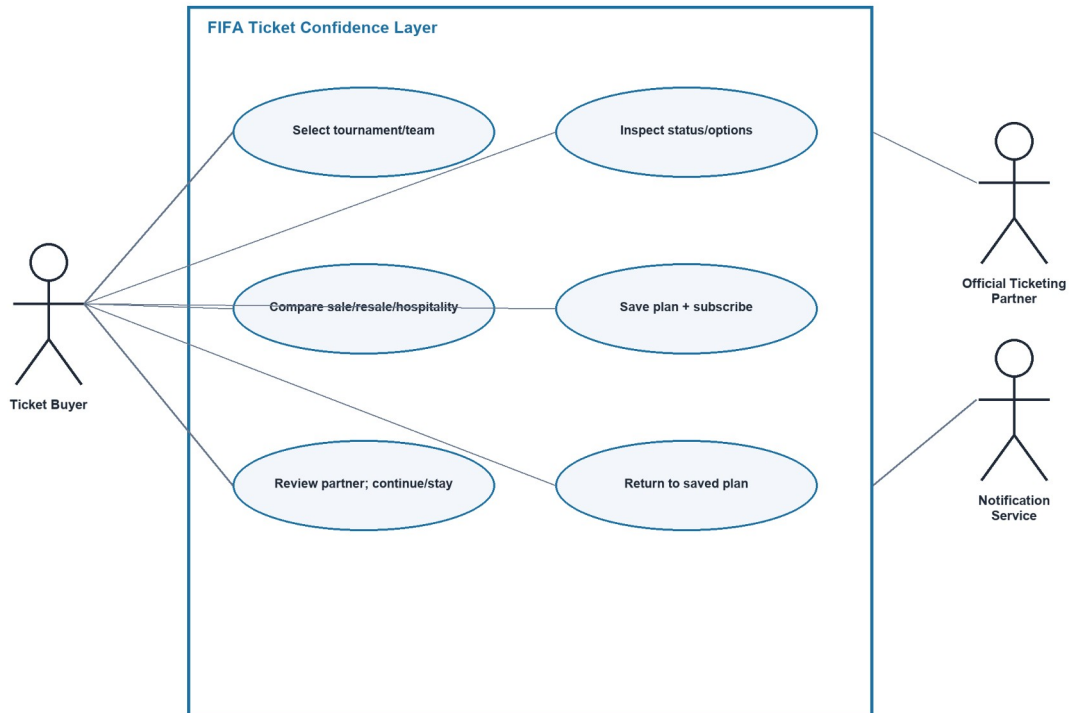
| Actor/system                   | Role                                                                                           | Trust boundary                                                               |
|--------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Ticket Buyer                   | Selects context, interprets governed status, saves a plan, and decides on an external handoff. | May cross partner, identity, and notification boundaries.                    |
| Official Ticketing Partner     | Provides the external official action.                                                         | External domain/account/payment context; purchase success is never inferred. |
| Notification/Identity Services | Persist optional plan/consent and deliver alerts.                                              | Personal data, consent, authentication, and retention.                       |
| Beginner / Returning Player    | Loads an artifact, requests review, practices, and chooses learning continuation.              | May be anonymous or authenticated.                                           |
| Analysis/Explanation Services  | Validate positions and supply analysis/explanation artifacts.                                  | Uncertainty, entitlement, and explanation quality.                           |
| Learning Catalog               | Supplies lessons/puzzles and entitlement metadata.                                             | Relevance, availability, progress, and return context.                       |

## 3. Concept-to-use-case mapping

| Concept | Primary use cases                      |
|---------|----------------------------------------|
| F-A1    | F-UC01, F-UC02, F-UC03, F-UC05, F-UC06 |
| F-A2    | F-UC01, F-UC02, F-UC05, F-UC06         |
| F-A3    | F-UC01, F-UC04, F-UC05, F-UC06         |
| C-A1    | C-UC01-C-UC06                          |
| C-A2    | C-UC01-C-UC04, C-UC06                  |
| C-A3    | C-UC01-C-UC03, C-UC05-C-UC06           |

## 4. FIFA UML use-case diagram

FIFA Ticket Confidence Layer - UML Use-Case Diagram



**Figure UC-F-SYS.** UML use-case diagram: named boundary, Ticket Buyer and external-system actors, use-case ellipses, and associations.

Source: *generated-diagrams/fifa-system-use-cases.png*

Related: F-P3; TP-FIFA

## 5. FIFA use case specifications

### F-UC01 Select tournament or team

| Field                      | Specification                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                                                         |
| Primary actor              | Ticket Buyer                                                                                                     |
| Supporting actors          | FIFA tournament catalog                                                                                          |
| Stakeholders and interests | Buyer wants correct context; FIFA content owner wants canonical tournament/team identity.                        |
| Trigger                    | The buyer opens the ticket-planning module with a tournament or team in mind.                                    |
| Preconditions              | The catalog has a governed identifier and display label; the module can disclose stale/unavailable catalog data. |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented.      |

| Field             | Specification                                                                 |
|-------------------|-------------------------------------------------------------------------------|
| Success guarantee | Selected ticket context and accessible confirmation.                          |
| Frequency         | At the start of a planning session; assumption to validate.                   |
| Priority          | High                                                                          |
| Links             | Proto-persona F-P3; TP-FIFA; concepts F-A1/F-A2/F-A3; evidence F2-E09, F2-E18 |
| Open issue        | Who owns catalog freshness across FIFA properties?                            |

### Main success scenario

1. Open the tournament/team selector.
2. Search or browse the governed catalog.
3. Inspect the matching label, competition, and season.
4. Choose one tournament or team.
5. Confirm the selected context shown in the planning header.
6. Continue to ticket-status inspection.

### Alternative and exception flows

- A1 at step 2: No result matches; show spelling guidance and browse categories without creating a fake item.
- A2 at step 4: Multiple seasons share a label; require an explicit season choice.
- E1 at step 2: Catalog is unavailable; retain the prior context and offer retry.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                            |
|---------------------|----------------------------------------------------------------------------------------|
| Required data       | Tournament/team ID, canonical label, competition, season, locale.                      |
| Rules               | Only governed catalog entries may be selected; ambiguous names require disambiguation. |
| Output artifact     | Selected ticket context and accessible confirmation.                                   |
| Special requirement | Search results expose type and season in text, not color alone.                        |

## F-UC02 Inspect official ticket status and available options

| Field                      | Specification                                                               |
|----------------------------|-----------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                    |
| Primary actor              | Ticket Buyer                                                                |
| Supporting actors          | Ticket Status Service                                                       |
| Stakeholders and interests | Buyer needs a bounded current state; data owner must avoid false precision. |
| Trigger                    | A ticket context is selected and the buyer asks what is                     |

| Field             | Specification                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------|
|                   | currently visible/available.                                                                                |
| Preconditions     | F-UC01 is complete; each returned field includes provenance and freshness or an unavailable marker.         |
| Minimal guarantee | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee | Readable status summary with provenance and one safe next action.                                           |
| Frequency         | Repeated during planning; frequency is an assumption, not measured.                                         |
| Priority          | Critical                                                                                                    |
| Links             | Proto-persona F-P3; TP-FIFA; concepts F-A1/F-A2; evidence F2-E09, F2-E18; F-D4                              |
| Open issue        | What official system is authoritative for each field?                                                       |

### Main success scenario

1. Request status for the selected context.
2. Read the plain-language overall state and last-updated time.
3. Review which official option categories have governed data.
4. Open the provenance explanation for a category.
5. Inspect the single next official action associated with that state.
6. Choose to compare options, save the plan, or stop.

### Alternative and exception flows

- A1 at step 2: Status is stale; label it stale and suppress action language that implies currency.
- A2 at step 3: A category has no governed data; show 'not provided' rather than 'unavailable'.
- E1 at step 1: Status service fails; show no cached value unless its timestamp and source remain visible.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                               |
|---------------------|-------------------------------------------------------------------------------------------|
| Required data       | Context ID, state label, option-category presence, source, timestamp, action eligibility. |
| Rules               | Unknown, not provided, closed, and sold out are distinct states.                          |
| Output artifact     | Readable status summary with provenance and one safe next action.                         |
| Special requirement | Timestamp is localized and screen-reader text conveys the same state as icons.            |

## F-UC03 Compare sale, resale, hospitality, and waiting states

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                                                    |
| Primary actor              | Ticket Buyer                                                                                                |
| Supporting actors          | Ticket Status Service; Hospitality Catalog                                                                  |
| Stakeholders and interests | Buyer wants comparable definitions; FIFA/partners need categories represented accurately.                   |
| Trigger                    | The buyer chooses Compare from a governed ticket-status summary.                                            |
| Preconditions              | At least one category has a governed value; unavailable fields are explicitly marked.                       |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Persisted comparison/filter state and selected option, if any.                                              |
| Frequency                  | Once or more per selected tournament; assumption to validate.                                               |
| Priority                   | High                                                                                                        |
| Links                      | Proto-persona F-P3; TP-FIFA; concepts F-A1; evidence F2-E09; F-D4                                           |
| Open issue                 | Which comparison fields are legally/operationally available per market?                                     |

### Main success scenario

1. Open the option comparison for the selected context.
2. Read category definitions for sale, resale, hospitality, and waiting.
3. Compare state, eligibility, key date, and official destination by row.
4. Filter rows to categories relevant to the buyer.
5. Inspect why any row lacks data or action.
6. Select one row for details or return to the unfiltered comparison.

### Alternative and exception flows

- A1 at step 2: A market does not support resale; explain the market rule instead of hiding the row.
- A2 at step 4: Filtering would produce zero rows; keep filters visible and offer Clear filters.
- E1 at step 3: Categories use incompatible freshness times; display each timestamp rather than a misleading shared timestamp.

### Use-case-specific data, rules, and special requirements

| Category      | Requirement                                                                              |
|---------------|------------------------------------------------------------------------------------------|
| Required data | Category taxonomy, state, eligibility summary, key date, destination, per-row timestamp. |

| Category            | Requirement                                                                |
|---------------------|----------------------------------------------------------------------------|
| Rules               | Rows cannot be ranked as better without an explicit buyer criterion.       |
| Output artifact     | Persisted comparison/filter state and selected option, if any.             |
| Special requirement | Table supports keyboard row navigation and a stacked mobile reading order. |

### F-UC04 Save a ticket plan and subscribe to verified alerts

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                                                    |
| Primary actor              | Ticket Buyer                                                                                                |
| Supporting actors          | Identity Service; Notification Service                                                                      |
| Stakeholders and interests | Buyer wants control and low alert noise; services need valid consent and deliverability.                    |
| Trigger                    | The buyer wants to avoid manually repeating the same check.                                                 |
| Preconditions              | A context exists; consent and authentication requirements are disclosed before collecting contact data.     |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Saved plan, subscription state, consent receipt, and controls.                                              |
| Frequency                  | Occasional setup followed by event-driven alerts; assumption to validate.                                   |
| Priority                   | High                                                                                                        |
| Links                      | Proto-persona F-P3; TP-FIFA; concepts F-A3; evidence F2-E09, F2-E18; F-D4                                   |
| Open issue                 | What retention period and channels are approved?                                                            |

#### Main success scenario

1. Open Save plan from the selected ticket context.
2. Choose tracked categories and optional team/date preferences.
3. Review which fields are stored locally or in an account.
4. Save the plan with a recognizable label.
5. Choose an alert channel and quiet-hours preference.
6. Confirm consent and view the verified-alert criteria.
7. Receive an accessible plan summary with edit/unsubscribe controls.



### Alternative and exception flows

- A1 at step 3: Buyer declines account storage; offer a local-only plan if supported.
- A2 at step 5: Buyer declines notifications; save the plan without a subscription.
- E1 at step 6: Channel verification fails; keep the plan and mark alerts inactive.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------|
| Required data       | Plan ID/label, context, categories, preferences, storage mode, channel, consent, quiet hours. |
| Rules               | No marketing consent is bundled with status alerts; alerts fire only on governed changes.     |
| Output artifact     | Saved plan, subscription state, consent receipt, and controls.                                |
| Special requirement | Consent is reversible; notification content minimizes sensitive preference data.              |

### F-UC05 Review partner identity and choose Continue or Stay

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                                                    |
| Primary actor              | Ticket Buyer                                                                                                |
| Supporting actors          | Official Ticketing Partner                                                                                  |
| Stakeholders and interests | Buyer needs informed control; FIFA and partner need a trustworthy boundary.                                 |
| Trigger                    | The buyer selects an action that leaves the FIFA-hosted module.                                             |
| Preconditions              | Destination is allow-listed and tied to the chosen option; the current plan/context can be preserved.       |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Handoff decision, retained context, and optional return token.                                              |
| Frequency                  | Before every external ticket/hospitality handoff.                                                           |
| Priority                   | Critical                                                                                                    |
| Links                      | Proto-persona F-P3; TP-FIFA; concepts F-A1/F-A2/F-A3; evidence F2-E10, F2-E11; F-D2                         |
| Open issue                 | Which partner/account disclosures are required for each route?                                              |

**Main success scenario**

1. Open the destination preview for the selected action.
2. Read the official partner name and destination domain.
3. Review whether sign-in, identity, or payment context may change.
4. Read what context will be preserved for return.
5. Choose Continue or Stay on FIFA.com.
6. If Continue, open the allow-listed destination and record a return token; if Stay, retain the current decision surface.

**Alternative and exception flows**

- A1 at step 2: Partner/domain verification is unavailable; disable Continue and offer retry.
- A2 at step 5: Buyer chooses Stay; close the preview without external navigation.
- E1 at step 6: Destination fails to open; keep the saved context and show a safe retry.

**Use-case-specific data, rules, and special requirements**

| Category            | Requirement                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------|
| Required data       | Partner ID/name, allow-listed domain, action type, account-boundary notice, return token.      |
| Rules               | No open redirect; Continue is unavailable when partner identity cannot be verified.            |
| Output artifact     | Handoff decision, retained context, and optional return token.                                 |
| Special requirement | Focus enters the preview, remains trapped appropriately, then returns to the invoking control. |

**F-UC06 Return to a saved plan after partner activity**

| Field                      | Specification                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------|
| Scope / level              | FIFA Ticket Confidence Layer / user goal                                                                         |
| Primary actor              | Ticket Buyer                                                                                                     |
| Supporting actors          | Official Ticketing Partner; Identity Service                                                                     |
| Stakeholders and interests | Buyer wants orientation; FIFA must not misrepresent partner-side completion.                                     |
| Trigger                    | The buyer follows a return link or reopens the planner after partner activity.                                   |
| Preconditions              | A valid local/account plan or time-limited return token exists; partner activity is not assumed to be available. |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented.      |
| Success guarantee          | Restored plan with explicit freshness and next actions.                                                          |
| Frequency                  | After an external handoff or later planning session; assumption to validate.                                     |

| Field      | Specification                                                               |
|------------|-----------------------------------------------------------------------------|
| Priority   | High                                                                        |
| Links      | Proto-persona F-P3; TP-FIFA; concepts F-A1/F-A2/F-A3; evidence F2-E11; F-D2 |
| Open issue | Can any partner return a signed completion state, and for how long?         |

### Main success scenario

1. Open the return link or Saved plans list.
2. Validate the return token or authenticate for the stored plan.
3. Restore tournament/team, comparison filters, and alert state.
4. Label the restored data with current freshness/provenance.
5. Show that partner-side purchase/activity status is unknown unless explicitly returned by a governed integration.
6. Let the buyer refresh, edit the plan, inspect status, or remove it.

### Alternative and exception flows

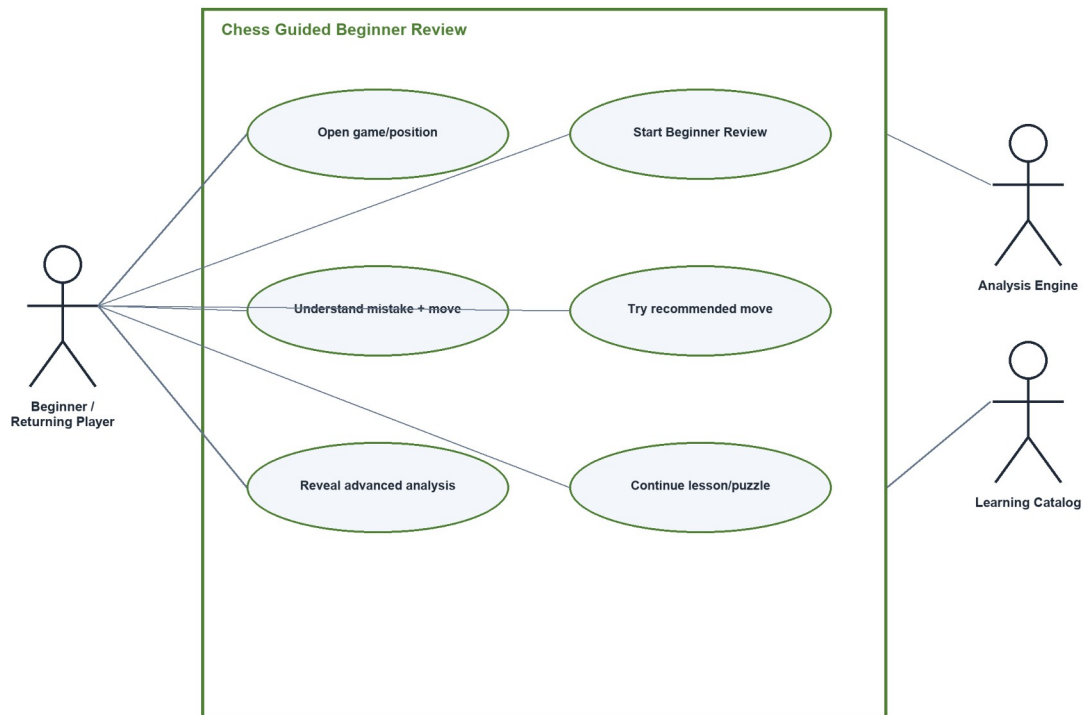
- A1 at step 2: Token expired; offer account lookup or safe plan recreation without exposing token data.
- A2 at step 3: Only a partial plan is available; list omitted fields before continuing.
- E1 at step 4: Refresh fails; keep the restored snapshot labeled with its prior timestamp.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Required data       | Return token, plan ID, saved context/filter/alerts, current status timestamp, optional governed partner result. |
| Rules               | Never infer purchase success from a return; expired tokens cannot be replayed.                                  |
| Output artifact     | Restored plan with explicit freshness and next actions.                                                         |
| Special requirement | Restoration is announced to assistive technology without moving focus unexpectedly.                             |

## 6. Chess UML use-case diagram

Chess Guided Beginner Review - UML Use-Case Diagram



**Figure UC-C-SYS.** UML use-case diagram: named boundary, player and external-system actors, use-case ellipses, and associations.

Source: generated-diagrams/chess-system-use-cases.png

Related: C-P1/C-P3; TP-CHESS

## 7. Chess use case specifications

### C-UC01 Open a completed game or analysis position

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                                    |
| Primary actor              | Beginner / Returning Player                                                                                 |
| Supporting actors          | Game Store; Analysis Engine                                                                                 |
| Stakeholders and interests | Player wants the intended position; platform needs valid, permission-appropriate game data.                 |
| Trigger                    | The player chooses a completed game, paste/import, or setup route for review.                               |
| Preconditions              | The source artifact is accessible; unsupported/invalid notation can be identified without losing input.     |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |

| Field             | Specification                                                                |
|-------------------|------------------------------------------------------------------------------|
| Success guarantee | Validated review artifact and entry state.                                   |
| Frequency         | Per review session; assumption to validate.                                  |
| Priority          | High                                                                         |
| Links             | Proto-persona C-P1; C-P3; TP-CHESS; concepts C-A1/C-A2/C-A3; evidence C2-E10 |
| Open issue        | Which anonymous/import formats are supported?                                |

### Main success scenario

1. Choose Completed game, Paste/import, or Set up position.
2. Select the game or enter the position data.
3. Review source, side to move, and orientation.
4. Correct any detected notation or setup issue.
5. Confirm the review artifact.
6. Open the review entry with Beginner Review and advanced-analysis choices.

### Alternative and exception flows

- A1 at step 1: No completed game exists; keep import and setup routes available.
- A2 at step 3: Orientation is wrong; switch sides before confirmation.
- E1 at step 2: Notation is invalid; identify the first invalid segment and preserve the remaining input.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------|
| Required data       | Game/position ID, notation, source, side to move, orientation, ownership/visibility.          |
| Rules               | Private games/positions follow account permissions; invalid data is never silently corrected. |
| Output artifact     | Validated review artifact and entry state.                                                    |
| Special requirement | Board setup has a text-notation alternative and fully labeled pieces/squares.                 |

## C-UC02 Start Beginner Review

| Field                      | Specification                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                         |
| Primary actor              | Beginner / Returning Player                                                                      |
| Supporting actors          | Analysis Engine                                                                                  |
| Stakeholders and interests | Player wants a low-vocabulary entry; engine owner must communicate uncertainty and availability. |
| Trigger                    | A valid game/position is open and the player chooses                                             |

| Field             | Specification                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------|
|                   | Beginner Review.                                                                                            |
| Preconditions     | C-UC01 is complete; the system can disclose analysis availability/entitlement.                              |
| Minimal guarantee | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee | Beginner Review overview or actionable unavailable state.                                                   |
| Frequency         | Per selected review artifact; assumption to validate.                                                       |
| Priority          | Critical                                                                                                    |
| Links             | Proto-persona C-P1; C-P3; TP-CHESS; concepts C-A1/C-A2/C-A3; evidence C2-E10; C-D2                          |
| Open issue        | What entitlement and expected-duration disclosures are accurate?                                            |

### Main success scenario

1. Choose Beginner Review from the review entry.
2. Read the mode summary: one critical point first, advanced detail optional.
3. Confirm whose perspective and skill-level language to use.
4. Start the requested analysis.
5. Wait with progress and cancellation controls.
6. Receive the Beginner Review overview or a clear unavailable explanation.

### Alternative and exception flows

- A1 at step 3: Player skips skill-level choice; use neutral beginner language, not a hidden profile inference.
- A2 at step 5: Player cancels; return to review entry with the artifact intact.
- E1 at step 4: Analysis is unavailable or restricted; explain the condition before suggesting another route.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                |
|---------------------|--------------------------------------------------------------------------------------------|
| Required data       | Artifact ID, perspective, language level, analysis request/status, entitlement disclosure. |
| Rules               | Beginner mode changes presentation, not engine truth; uncertainty is retained.             |
| Output artifact     | Beginner Review overview or actionable unavailable state.                                  |
| Special requirement | Progress is perceivable without relying on animation; cancellation is keyboard accessible. |

### C-UC03 Understand the main mistake and recommended move

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                                    |
| Primary actor              | Beginner Player                                                                                             |
| Supporting actors          | Analysis Engine; Explanation Service                                                                        |
| Stakeholders and interests | Beginner wants meaning, not just a label;<br>engine/explanation owners need calibrated language.            |
| Trigger                    | Beginner Review returns at least one explainable critical position.                                         |
| Preconditions              | C-UC02 succeeded; a move recommendation and confidence/uncertainty representation exist.                    |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Comprehended-position view and chosen next action.                                                          |
| Frequency                  | One primary explanation per beginner overview;<br>assumption to validate.                                   |
| Priority                   | Critical                                                                                                    |
| Links                      | Proto-persona C-P1; TP-CHESS; concepts C-A1/C-A2/C-A3; evidence C2-E08, C2-E10; C-D2                        |
| Open issue                 | Which explanation rules are safe at each language level?                                                    |

#### Main success scenario

1. Open the primary critical-position card.
2. See the played move and recommended move in notation and on the board.
3. Read the immediate tactical/positional consequence in plain language.
4. Inspect the evidence cue or uncertainty note behind the recommendation.
5. Move between before and after board states.
6. Choose Try this move, Next critical point, or More analysis.

#### Alternative and exception flows

- A1 at step 1: No single critical point is reliable; present a neutral summary and explain why.
- A2 at step 3: A term is unfamiliar; open an inline definition without leaving the position.
- E1 at step 4: Explanation cannot be generated reliably; show move comparison without synthetic prose.

#### Use-case-specific data, rules, and special requirements

| Category      | Requirement                                                                               |
|---------------|-------------------------------------------------------------------------------------------|
| Required data | Position, played/recommended moves, evaluation change, explanation concepts, uncertainty. |
| Rules         | Do not state one move as universally best without the                                     |

| Category            | Requirement                                                                      |
|---------------------|----------------------------------------------------------------------------------|
|                     | engine context; avoid invented player intent.                                    |
| Output artifact     | Comprehended-position view and chosen next action.                               |
| Special requirement | Every visual arrow/color has text notation and ordered board-state descriptions. |

### C-UC04 Try the recommended move

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                                    |
| Primary actor              | Beginner Player                                                                                             |
| Supporting actors          | Move Validator; Analysis Engine                                                                             |
| Stakeholders and interests | Player wants immediate practice; validator must distinguish legal, recommended, and alternative moves.      |
| Trigger                    | The player selects Try this move from a critical-position explanation.                                      |
| Preconditions              | A legal starting position and recommended move are available; practice mode is non-rated.                   |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Practice attempt, feedback, and retry/continue choice.                                                      |
| Frequency                  | Zero or more times per critical position.                                                                   |
| Priority                   | High                                                                                                        |
| Links                      | Proto-persona C-P1; TP-CHESS; concepts C-A1/C-A2; evidence C2-E08, C2-E10                                   |
| Open issue                 | How many alternatives receive tailored feedback?                                                            |

#### Main success scenario

1. Open the critical position in practice mode.
2. Read the goal and whose turn it is.
3. Enter a move by board or notation.
4. Receive legal/illegal validation without revealing the answer prematurely.
5. If legal, compare the move with the recommendation and see the key consequence.
6. Retry, reveal the recommended move, or continue to practice.

#### Alternative and exception flows

- A1 at step 3: Player requests a hint; reveal one bounded cue and record no failure judgment.
- A2 at step 5: Legal alternative is reasonable; explain the difference instead of marking it simply wrong.
- E1 at step 4: Input method fails; preserve the position and offer notation entry/reset.



### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------|
| Required data       | Practice position, entered move, legality, recommended/acceptable alternatives, hint state.     |
| Rules               | Practice never changes rating/game history; legality and recommendation are separate judgments. |
| Output artifact     | Practice attempt, feedback, and retry/continue choice.                                          |
| Special requirement | No time limit; keyboard move entry and full board reset are available.                          |

### C-UC05 Reveal advanced analysis on demand

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                                    |
| Primary actor              | Returning Player                                                                                            |
| Supporting actors          | Analysis Engine                                                                                             |
| Stakeholders and interests | Returning player wants depth; beginner flow must remain stable and recoverable.                             |
| Trigger                    | The player chooses More analysis from a beginner explanation or overview.                                   |
| Preconditions              | A review artifact exists; advanced data availability and terminology level are disclosed.                   |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Advanced view state plus intact return location.                                                            |
| Frequency                  | Optional per position; assumption to validate.                                                              |
| Priority                   | Medium                                                                                                      |
| Links                      | Proto-persona C-P3; TP-CHESS; concepts C-A1/C-A3; evidence C2-E10                                           |
| Open issue                 | Which advanced controls belong in the first disclosure tier?                                                |

### Main success scenario

1. Activate the labeled More analysis disclosure.
2. Read a short preview of the additional data types.
3. Expand engine lines, evaluations, or setup controls intentionally.
4. Adjust one advanced setting if needed.
5. Inspect the updated advanced result with provenance.
6. Collapse advanced analysis and return to the same beginner position.

### Alternative and exception flows

- A1 at step 3: Advanced data is unavailable; leave the beginner explanation visible and state why.
- A2 at step 4: A setting changes the evaluated position; require confirmation before replacing the current context.
- E1 at step 5: Engine refresh fails; retain the prior result with its timestamp/status.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------|
| Required data       | Disclosure state, engine lines/evaluations, settings, provenance, refresh status.             |
| Rules               | Advanced detail is opt-in and reversible; collapsing never discards the beginner explanation. |
| Output artifact     | Advanced view state plus intact return location.                                              |
| Special requirement | Disclosure communicates expanded/collapsed state programmatically and restores focus.         |

### C-UC06 Continue with a relevant lesson or puzzle

| Field                      | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Scope / level              | Chess Guided Beginner Review / user goal                                                                    |
| Primary actor              | Beginner / Returning Player                                                                                 |
| Supporting actors          | Learning Catalog                                                                                            |
| Stakeholders and interests | Player wants relevant continuation; catalog owner needs accurate topic/level/entitlement metadata.          |
| Trigger                    | The player finishes an explanation/practice attempt and chooses a learning follow-up.                       |
| Preconditions              | A concept tag can be mapped to governed catalog items; entitlement is known or disclosed.                   |
| Minimal guarantee          | The originating artifact/context remains recoverable; no unsupported state or external success is invented. |
| Success guarantee          | Opened learning item, saved return context, and optional progress link.                                     |
| Frequency                  | Optional at the end of a review segment.                                                                    |
| Priority                   | High                                                                                                        |
| Links                      | Proto-persona C-P1; C-P3; TP-CHESS; concepts C-A1/C-A2/C-A3; evidence C2-E07, C2-E08, C2-E15                |
| Open issue                 | How are relevance and duration validated for beginners?                                                     |

### Main success scenario

1. Open the Recommended next step panel.

2. Read why each lesson or puzzle relates to the reviewed position.
3. Compare type, level, duration estimate, and entitlement.
4. Choose one lesson or puzzle.
5. Confirm whether progress will be saved and where the review return link appears.
6. Open the learning item with the review context preserved.
7. Complete/exit and return to the review or learning path.

### Alternative and exception flows

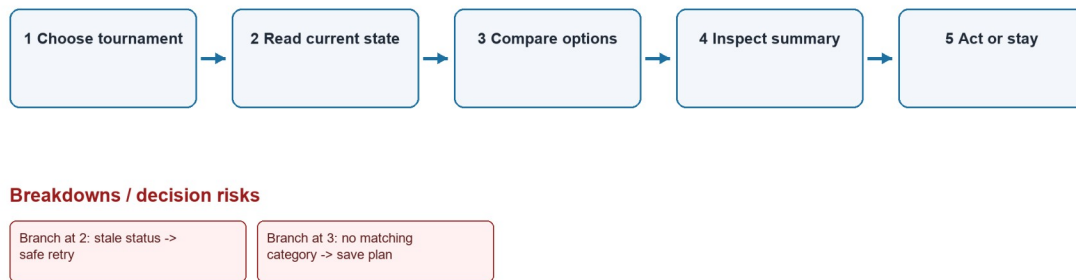
- A1 at step 2: No strong match exists; say so and offer browse by concept instead of a false recommendation.
- A2 at step 3: Item is restricted; disclose entitlement before selection and keep an accessible alternative.
- E1 at step 6: Catalog item fails to open; retain review context and offer another item.

### Use-case-specific data, rules, and special requirements

| Category            | Requirement                                                                               |
|---------------------|-------------------------------------------------------------------------------------------|
| Required data       | Concept tags, item ID/type/level/duration, relevance reason, entitlement, return context. |
| Rules               | Recommendation rationale is visible; restricted items are not the only continuation path. |
| Output artifact     | Opened learning item, saved return context, and optional progress link.                   |
| Special requirement | Cards have descriptive link names and predictable return navigation.                      |

## 8. Alternative-concept flow models

### F-A1 Status Dashboard Flow

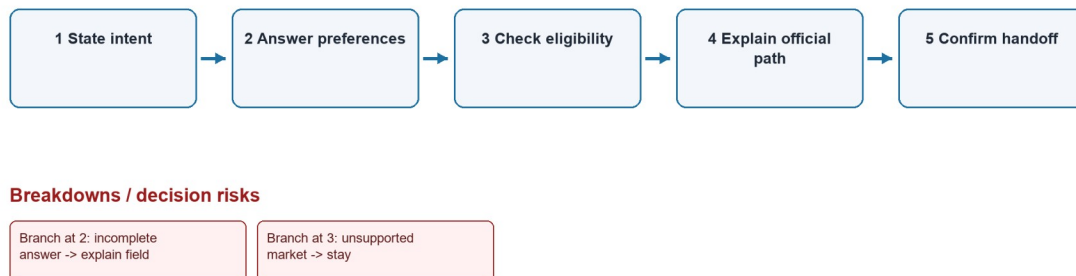


**Figure UC-F-A1.** Status Dashboard flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: *generated-diagrams/f-a1-flow.png*

Related: F-P3; TP-FIFA

### F-A2 Guided Ticket Concierge Flow

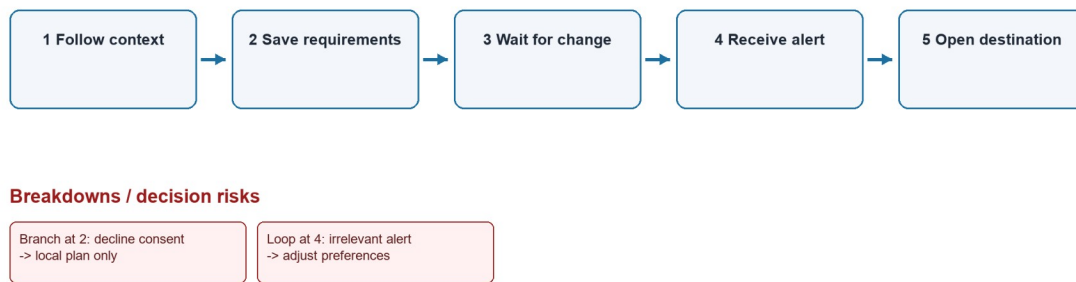


**Figure UC-F-A2.** Guided Ticket Concierge flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: *generated-diagrams/f-a2-flow.png*

Related: F-P3; TP-FIFA

### F-A3 Alert-First Ticket Planner Flow

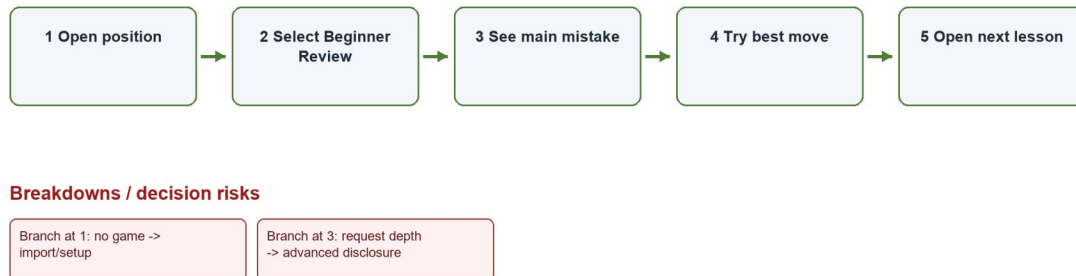


**Figure UC-F-A3.** Alert-First Ticket Planner flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: *generated-diagrams/f-a3-flow.png*

Related: F-P3; TP-FIFA

### C-A1 Beginner Analysis Preset Flow

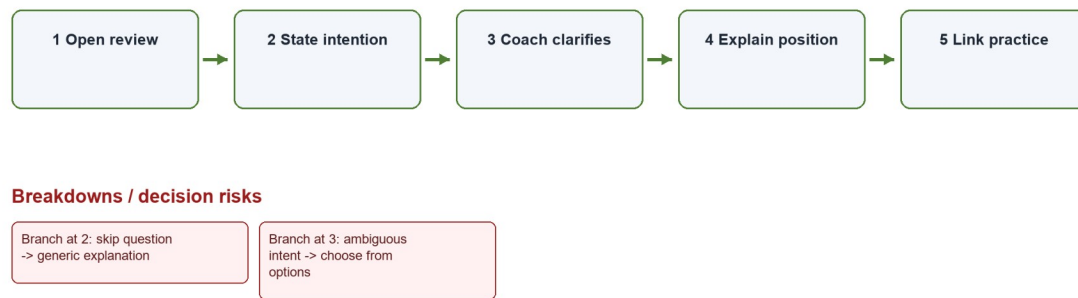


**Figure UC-C-A1.** Beginner Analysis Preset flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: *generated-diagrams/c-a1-flow.png*

Related: C-P1/C-P3; TP-CHESS

### C-A2 Conversational Coach Flow

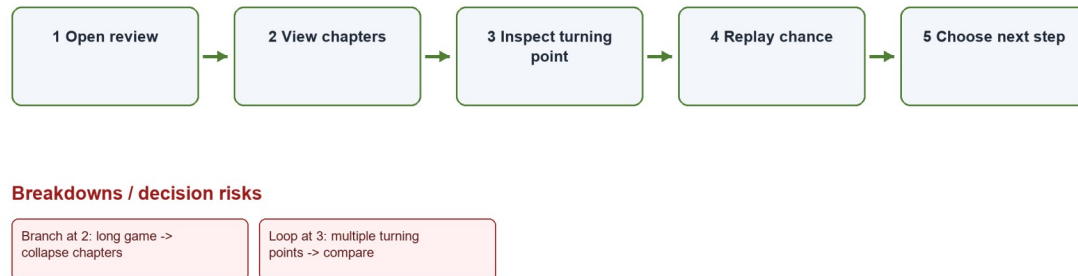


**Figure UC-C-A2.** Conversational Coach flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: generated-diagrams/c-a2-flow.png

Related: C-P1/C-P3; TP-CHESS

### C-A3 Visual Game Story Flow



**Figure UC-C-A3.** Visual Game Story flow model with numbered exact steps and labeled branches/breakdowns; it is not a use-case diagram.

Source: generated-diagrams/c-a3-flow.png

Related: C-P1/C-P3; TP-CHESS

## 9. Shared system-level nonfunctional requirements

| Category  | System-level requirement                               |
|-----------|--------------------------------------------------------|
| Usability | Plain task language, visible system status, low-memory |

| Category                    | System-level requirement                                                                                                                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | choices, explicit closure, and recoverable reversal (LN02 - Fundamental Concepts, p. 116).                                                                |
| Accessibility               | Keyboard operation, visible focus, logical reading order, text alternatives, labeled controls/states/errors, no color-only meaning, and scalable layouts. |
| Security / trust            | Allow-listed external destinations, provenance, calibrated uncertainty, protected tokens/preferences, and no inferred external success.                   |
| Privacy                     | Data minimization, purpose-specific consent, retention disclosure, edit/delete controls, and no bundled marketing consent.                                |
| Data freshness / provenance | Every governed state exposes source and timestamp; stale, unknown, not provided, closed, and unavailable remain distinct.                                 |
| Error recovery              | Preserve valid input/context, focus the exact error, provide retry/safe exit, and avoid destructive silent correction.                                    |
| Localization                | Localize dates/times/language while preserving canonical identifiers, chess notation, partner domains, and status semantics.                              |

## 10. Traceability and open issues

| Tough problem | Concepts       | Use cases     | Evidence                                                  |
|---------------|----------------|---------------|-----------------------------------------------------------|
| TP-FIFA       | F-A1/F-A2/F-A3 | F-UC01-F-UC06 | F2-E09-E11/E18; F-D2/F-D4                                 |
| TP-CHESS      | C-A1/C-A2/C-A3 | C-UC01-C-UC06 | C2-E07/E08/E10/E15; C-D1/C-D2 (entry-only PA2 limitation) |

- Which official system owns each ticket state and freshness field?
- Which markets and partner disclosures must each FIFA path express?
- Which explanations can be produced reliably with calibrated uncertainty?
- Which entitlement alternatives and return contexts are supported?
- Which PA3 measures will validate the stated hypotheses without treating them as prior results?